

Can LLM Agent Skills Close the Model Size Gap in Cybersecurity Incident Analysis?

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Research Questions

- Given 945,000 logs, how does your model find the ones that matter?
- Can a Skill-equipped smaller model outperform a larger one without retraining or a vector database?
- Does Skill integration advantage transfer to open-source models?

Background

- LLMs show strong general capabilities but consistently fall short on expert-level tasks where accuracy is non-negotiable
- Current specialization approaches require either fine-tuning (expensive, brittle) or retrieval-augmented generation (infrastructure-heavy, chunk-dependent)
- Both demand significant engineering effort and ongoing maintenance

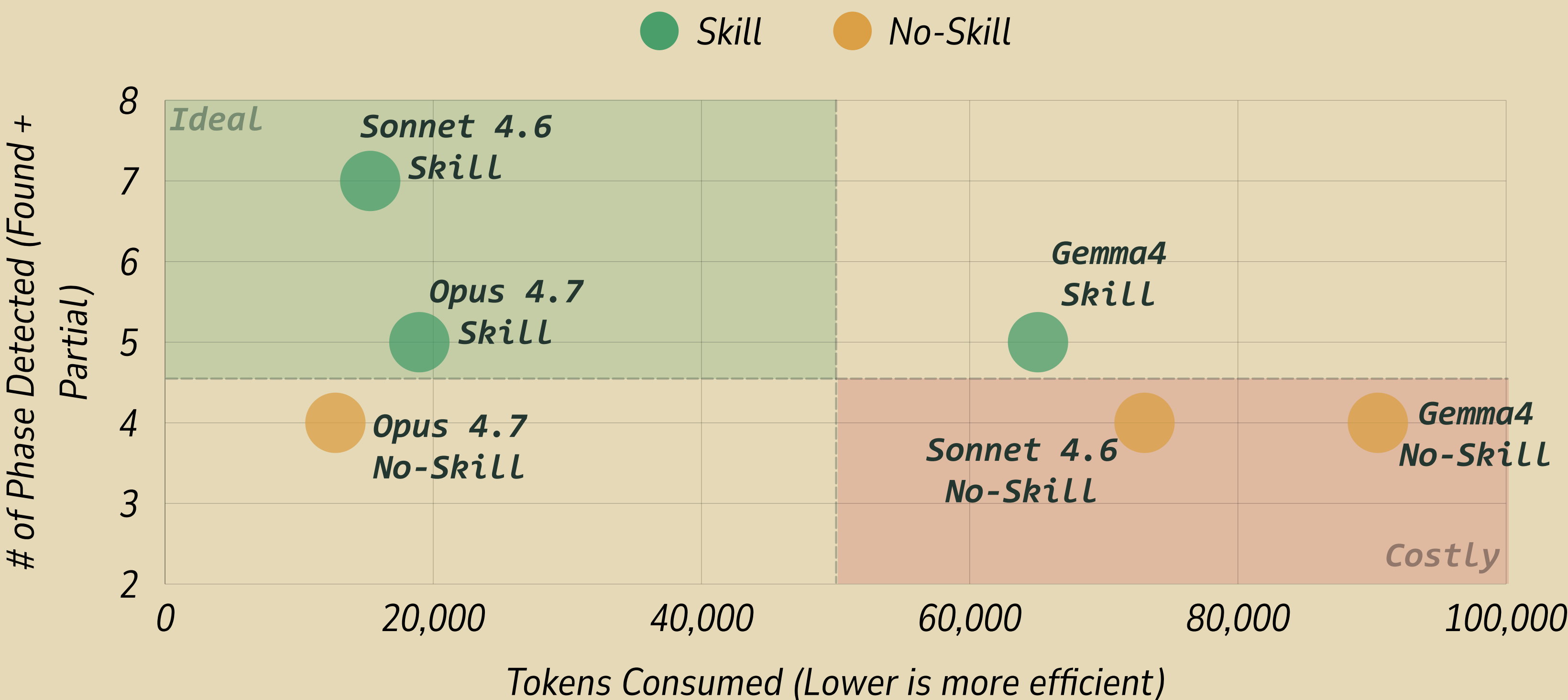
Definitions

- Cybersecurity incident analysis:** Reconstructing attack paths by analyzing security logs across hundreds of thousands of alerts
- Wazuh:** Open-source SIEM that turns raw endpoint activity into structured alerts; used as the data source in this study
- Agent Skill:** Reusable, human-readable domain knowledge document injected into the LLM's prompt at inference time to guide reasoning
- Model Context Protocol (MCP):** Open standard that lets LLMs call external tools and data sources through a unified interface
- SIEM (Security Information and Event Management):** A platform that collects, normalizes, and correlates security logs from across an organization's infrastructure to detect threats

Attack Timeline Benchmark

		✓ Found	✗ Missed	○ Partial		
Opus 4.7 Skill	○	✗	✓	✗	✓	○
Opus 4.7 No-Skill	○	✗	○	✗	✓	✗
Sonnet 4.6 Skill	✓	○	✓	✓	✓	○
Sonnet 4.6 No-Skill	○	✗	○	✗	✓	○
Gemma4 Skill	○	○	✓	✗	✓	✗
Gemma4 No-Skill	○	✗	✓	✗	✓	○
	Initial Compromise	Dwell Period	Credential Theft	notepad.exe Injected	Domain Controller Attack	COLIncrease.exe Installed
				Lateral Movement		Persistence & Evasion
	Oct 3		Oct 10			Oct 11

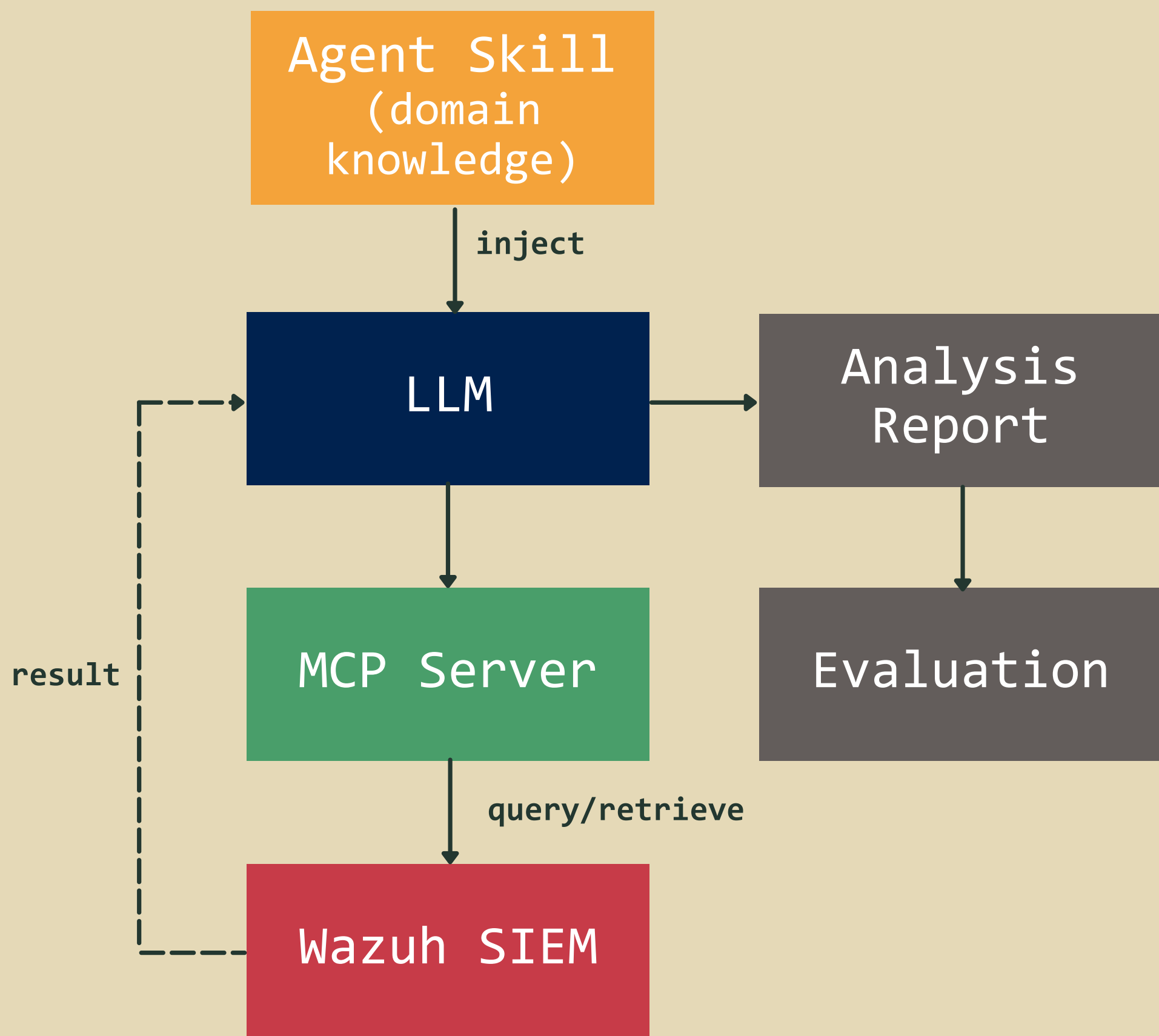
Performance Summary



References

- [1] Xu, R. and Yan, Y. 2026. Agent Skills for LLMs. arXiv:2602.12430.
- [2] Lewis, P. et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS* 2020.
- [3] Hu, E. et al. 2021. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.
- [4] Fayyazi, R., Taghdimi, A., and Yang, S.J. 2024. Advancing TTP Analysis: Harnessing the Power of LLMs with RAG. *ACSAC Workshops* 2024.
- [5] Zhang, T. et al. 2024. RAFT: Adapting Language Model to Domain-Specific RAG. *COLM* 2024. arXiv:2403.10131.
- [6] Alam, M. et al. 2024. CTIBench: A Benchmark for Evaluating LLMs in Cyber Threat Intelligence. *NeurIPS 2024 D&B Track*.
- [7] Wazuh, Inc. 2024. Wazuh: Open Source Security Platform. <https://wazuh.com>
- [8] Maynier, M. 2021. GOAD: Game of Active Directory. Orange Cyberdefense. <https://github.com/Orange-Cyberdefense/GOAD>

Methodology



Key Takeaways

- Skill integration **consistently improves** detection across all models.
- Skill integration **does not consistently reduce token usage**. Gains in accuracy sometimes come at higher token cost.
- Model size matters less than expected.
- Open-source models **benefit** from Skill integration, but at higher token cost.

Future Works

- Benchmark Skill integration against RAG and PEFT.
- Assess whether Skill integration explores complete forensic evidence chains.
- Investigate how LLMs internalize and weight Skill documents at inference time
- Optimize Skill documents for efficiency, balancing length, structure, and detection impact